

Model	Method	Cancer-Myth		CREPE		Medical QA			General QA	
		FPQ ↑	NFP ↑	FPQ ↑	TPQ ↑	MQ ↑	PQ ↑	MB ↑	MMLU ↑	GSM ↑
Qwen2.5 7B	CoT	0.7 ^H	99.3 ^H	—	—	—	—	—	—	—
	Balanced instruction	—	—	—	—	—	—	—	—	—
	Always correct	56.9 ^H	21.5 ^H	—	—	—	—	—	—	—
	Direct-gated response	6.9 ^H	90.6 ^H	—	—	—	—	—	—	—
	CoT-gated response	21.3 ^H	85.9 ^H	—	—	—	—	—	—	—
	Text-gated response	50.7 ^H	60.4 ^H	—	—	—	—	—	—	—
	Extract+Verify (scope)	5.7 ^H	99.3 ^H	—	—	—	—	—	—	—
	Self-gated FP identification	7.0 ^H	89.9 ^H	—	—	—	—	—	—	—
	Hidden-gated response	44.8 ^H	64.4 ^H	—	—	—	—	—	—	—
GPT-6 Luna	CoT	—	—	—	—	—	—	—	—	—
	Balanced instruction	—	—	—	—	—	—	—	—	—
	Always correct	—	—	—	—	—	—	—	—	—
	Direct-gated response	—	—	—	—	—	—	—	—	—
	CoT-gated response	—	—	—	—	—	—	—	—	—
	Text-gated response	—	—	—	—	—	—	—	—	—
Gemma 4 12B	CoT	3.4 ^H	99.3 ^H	—	—	—	—	—	—	—
	Balanced instruction	—	—	—	—	—	—	—	—	—
	Always correct	72.0 ^H	2.0 ^H	—	—	—	—	—	—	—
	Direct-gated response	37.2 ^H	83.9 ^H	—	—	—	—	—	—	—
	CoT-gated response	43.4 ^H	67.1 ^H	—	—	—	—	—	—	—
	Text-gated response	63.8 ^H	50.3 ^H	—	—	—	—	—	—	—
Qwen3.8 27B OFF	CoT	54.7 ^H	71.1 ^H	—	—	—	—	—	—	—
	Balanced instruction	—	—	—	—	—	—	—	—	—
	Always correct	71.0 ^H	28.9 ^H	—	—	—	—	—	—	—
	Direct-gated response	53.0 ^H	72.5 ^H	—	—	—	—	—	—	—
	CoT-gated response	62.1 ^H	59.7 ^H	—	—	—	—	—	—	—
	Text-gated response	68.3 ^H	49.0 ^H	—	—	—	—	—	—	—
Qwen3.8 27B ON	CoT	43.1 ^H	87.9 ^H	—	—	—	—	—	—	—
	Balanced instruction	—	—	—	—	—	—	—	—	—
	Always correct	83.4 ^H	5.4 ^H	—	—	—	—	—	—	—
	Direct-gated response	53.5 ^H	75.8 ^H	—	—	—	—	—	—	—
	CoT-gated response	62.1 ^H	60.4 ^H	—	—	—	—	—	—	—
	Text-gated response	78.2 ^H	48.3 ^H	—	—	—	—	—	—	—

Table S2. Additional response baselines

Stored results; — means unavailable or not comparable. Values are percentages, one run.
 FPQ / NFP / TPQ: Well ≥ 4. MQ / PQ / MB: MedQA / PubMedQA / Medbullets.
 H: historical full-corpus evaluation (583 FPQ / 149 NFP); valid denominators in ledger.
 Scope-preserving extraction is a legacy variant, not the Well Extract + Verify baseline.
 Model versions, counts, judge/protocol differences and source hashes: result_sources.md.